The steeper 1/f slope of MEG power spectrum is associated with stronger gamma synchrony, measured via inter-trial phase coherence (ITPC) in the 40-Hz auditory steady-state response

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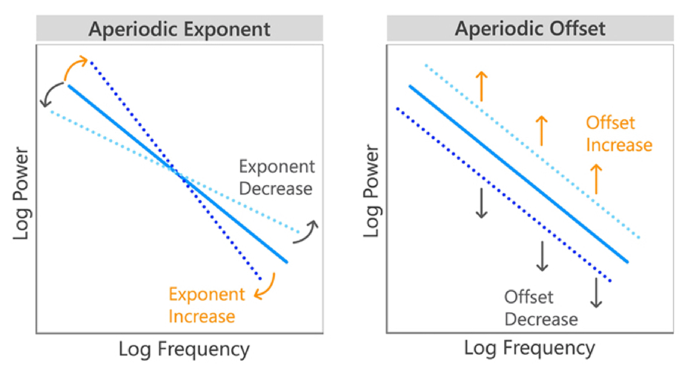
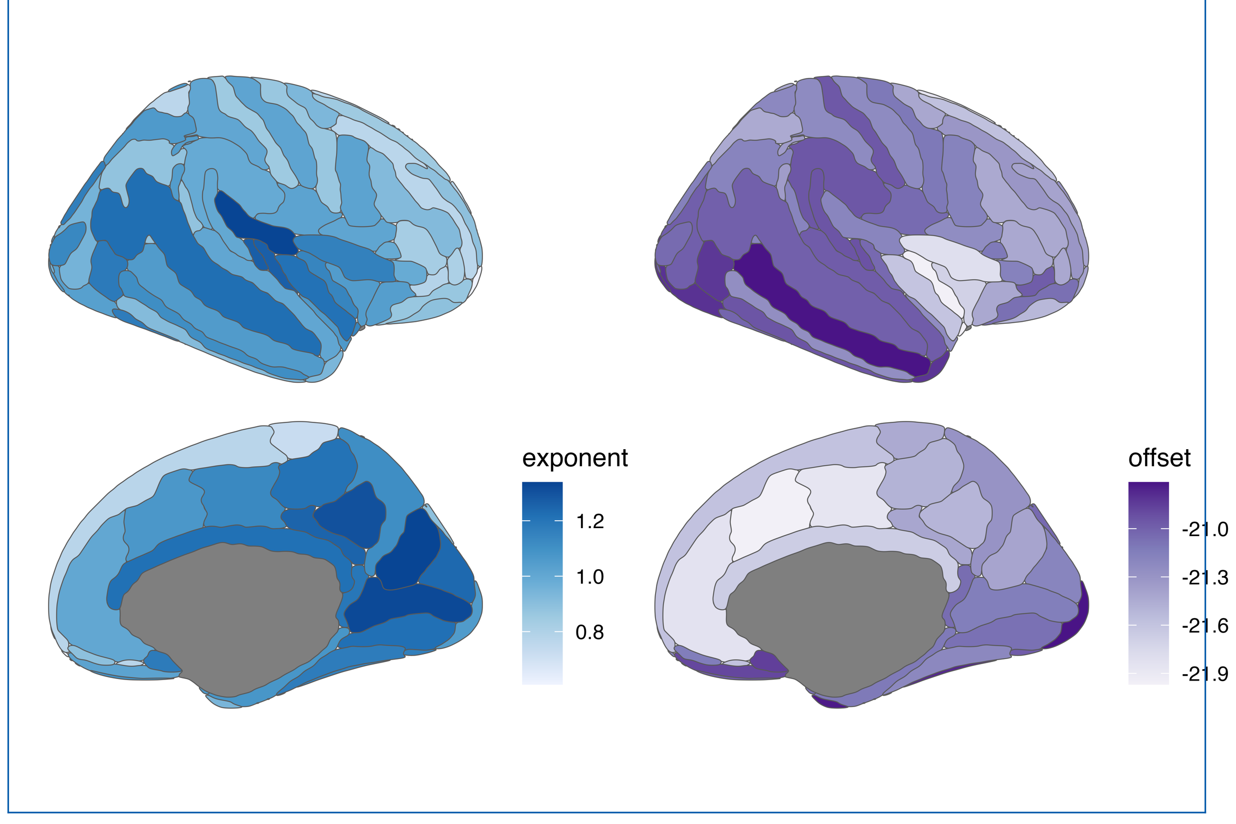
Hypothesis. We explored whether a steeper slope of the 1/f PSD component (i.e., a larger spectral exponent) in MEG signals is associated with higher auditory steady-state response (ASSR) synchrony, as measured by inter-trial phase coherence (ITPC) during 40 Hz auditory stimulation.

Introduction

- Power spectral density (PSD) of electrophysiological activity may be parametrized by oscillatory components and aperiodic component, where the power of the signal decay with a 1/f power law
- The timescale and magnitude of this aperiodic activity can be described by the slope and offset of this 1/f decay [2]
- Simplified formula describing 1/f component of the PSD of the signal: $L(f) = b - \log(k + f^{\chi})$; The exponent χ is related to the negative slope of the power spectrum in log-log space, that is, the inverse relationship between power and frequency; The offset b describe the uniform shift in power across frequencies
- These parameters are most likely governed by waxing and waning of oscillatory components, and can be associated to biological mechanisms (e.g. excitation:inhibition balance as summed by [7], cognitive processes, and group differences).
- Graphical relationship of spectral power and offset and exponent (from [3])

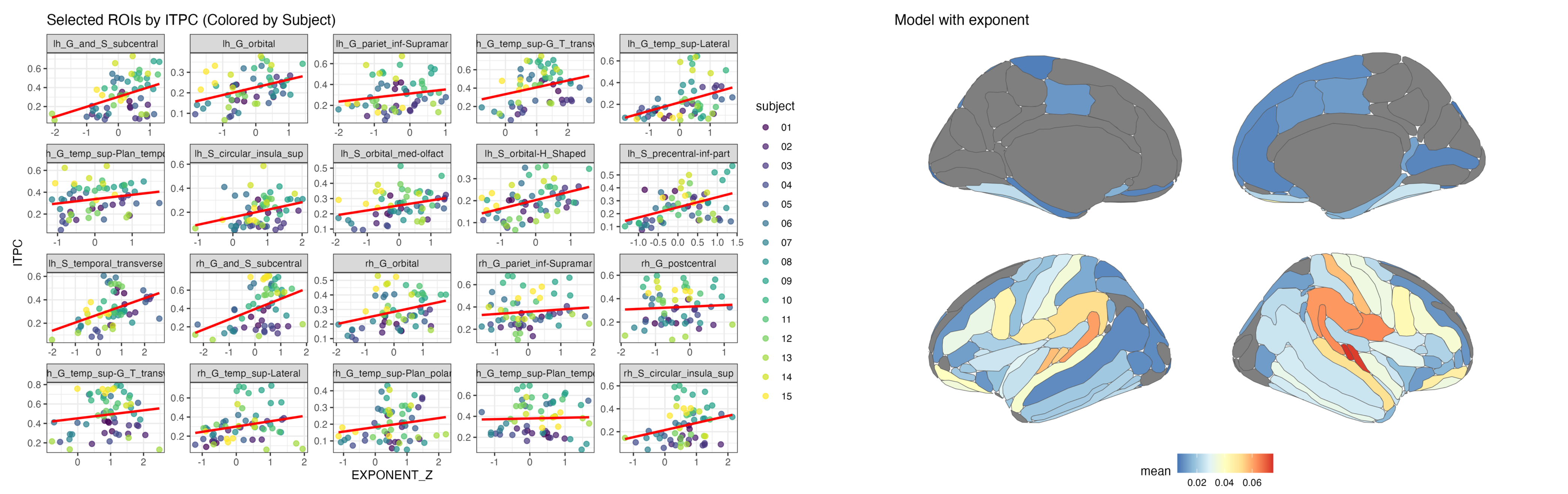


1 **Spectral Parametrization.** Mean values of estimated exponent and offset describing the 1/f part of spectral power across different regions (right hemisphere).



- Synchronization of brain activity in the gamma frequency is a hallmark of brain function, and an indicator of malfunctioning[4].
- Auditory stimulation in the gamma range is a reliable way to experimentally investigate this phenomenon, looking at auditory steady-state responses (ASSR), the effect of entrainment of brain activity to the frequency and phase of external auditory stimuli.
- ASSR are usually investigated by looking at the inter-trial phase coherence (ITPC) [4].

2 **Relationship between the slope (parametrized by exponent and the estimated ITPC).** Figure on the left: Relationship between Spectral Exponent (z-scored) and ITPC across selected ROIs. Red line indicates linear fit. Data points colored by subject.



Data and Methods

Data:

- **n=15 healthy participants;** In block 180 trials of 40-Hz auditory stimulation (1s duration) [4]
- Source-reconstructed MEG data from 148 cortical regions (Destrieux atlas)

Spectral Parameterization:

- PSD computed for 1s pre-stimulus and stimulus periods per trial/region
- Exponent and offset extracted using `specparam` [2] from 3-45 Hz range

Phase-Locking Measurement:

- ITPC computed via Morlet wavelets at 40 Hz [4]
- Baseline: $[-500, -200]$ ms; Stimulation: $[400, 700]$ ms post-onset

Statistical Analysis:

- **Bayesian multilevel modeling** via Region-Based Analysis (RBA) [1], accounts for spatial structure and multiple comparisons through hierarchical pooling
- Posterior distributions quantify effect uncertainty across 148 regions

Models tested:

1. ITPC ∼ Stimulus (entrainment effect)
2. Exponent ∼ Stimulus (stability test)
3. Offset ∼ Stimulus

Primary model (exponent-ITPC relationship):

$$\text{ITPC}_{\text{stim}} \sim 1 + \text{exponent}_z + (1 + \text{exponent}_z \mid \text{subject}) + (1 \mid \text{ROI})$$

where exponent$_z$ is within-subject z-scored. Fitted using `brms` package.

Results

Spatial Distribution & Stability:

- Exponent and offset varied systematically across brain regions (Box 1)
- Exponent remained stable during auditory stimulation (no evidence for Stimulus effect)
- Offset showed no systematic changes across conditions

ITPC Response to Auditory Stimulation:

- Strong stimulus effect on ITPC (baseline vs. stimulation) in auditory cortex, temporal lobe, inferior sensorimotor regions, and inferior frontal regions
- Right hemisphere showed somewhat stronger effects

Exponent-ITPC Relationship:

- Positive relationship between exponent and ITPC during stimulation
- Steeper spectral slopes associated with stronger phase-locking
- Regional specificity: Strongest effects in right superior temporal gyrus, temporal pole, insula, and inferior frontal regions—overlapping with regions showing robust ASSR responses
- Offset showed no significant relationship with ITPC

Statistical Approach:

- Bayesian multilevel modeling accounts for spatial structure and multiple comparisons through hierarchical pooling
- Posterior distributions and P^{+} quantify uncertainty in estimates

Conclusion

Main Finding:

- Steeper spectral slopes associated with stronger gamma phase-locking, particularly in auditory and temporal regions showing robust ASSR responses

Two Possible Mechanisms:

- **Biological:** Both measures may reflect E/I balance—stronger inhibitory tone (steeper slope) provides temporal scaffolding for precise synchronization [7]
- **Methodological:** Spectral slope influences phase estimation reliability and SNR [6, 5]
- Regional specificity and exponent stability suggest biological contribution, though methodological confounding cannot be fully ruled out

Implications:

- Exploratory findings ($n = 15$) warrant replication in larger samples
- Combined spectral-oscillatory measures may enhance sensitivity to circuit dysfunction
- Future work should distinguish biological from methodological contributions through controlled simulations and clinical validation

References

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